r/DefendingAIArt

Your Daily Activities Use More Water Than AI

Posted by u/Mikhael_Love • • 0 points • 24 comments

[Water Pouring into Server Room Glass - AI Generated Image by Mikhael Love in the style of Photography](https://preview.redd.it/rwt5wdyhj3if1.png?width=1728&format=png&auto=webp&s=61027ab486e88cf516cf5881ce66615d5d1ad0ff)

>***I noticed recently the topic of water usage has come up again so I thought I'd share this blog post. The data should still be accurate.***

I keep seeing these posts about AI water usage that make it sound like ChatGPT is sucking our planet dry. The numbers they throw around sound pretty scary at first. You'll see reports claiming ChatGPT uses around 5ml of water per query [1](https://www.seangoedecke.com/water-impact-of-ai/), then another article says it's actually 30ml for a single question [2](https://nationalcentreforai.jiscinvolve.org/wp/2025/05/02/artificial-intelligence-and-the-environment-putting-the-numbers-into-perspective/).

But here's the thing – when you actually look at the numbers, it gets way more interesting. Yeah, training something like GPT-3 can use between 2 and 15 million liters of water [1](https://www.seangoedecke.com/water-impact-of-ai/). That sounds massive. But your actual daily ChatGPT usage? It's pretty tiny compared to stuff you do without thinking twice about it.

If you ask ChatGPT five questions a day, you're looking at about 150ml of water[2](https://nationalcentreforai.jiscinvolve.org/wp/2025/05/02/artificial-intelligence-and-the-environment-putting-the-numbers-into-perspective/). That's less than a small cup of coffee.

So I'm going to break down what these numbers actually mean – em dashes included – and compare them to things you probably do every day. You might be suprised to find out that your shower uses way more water than all your AI conversations combined. I'll also explain why some of these numbers you see in headlines can be pretty misleading, and what we should really be paying attention to when we talk about AI's environmental impact.

How much water does AI really use?

When researchers started digging into AI's water footprint, the numbers were all over the place. Let me walk you through what the actual data shows.

The origin of the 500ml claim

That “500ml bottle” number everyone talks about came from a 2023 paper by Li et al. Basically, they suggested GPT-3 uses around 500ml of water for every 10-50 medium-length responses [3](https://generative-ai-newsroom.com/the-often-overlooked-water-footprint-of-ai-models-46991e3094b6). But here’s where it gets intresting – location matters a lot. In Arizona, you might hit 500ml after just 17 responses, while in Ireland it would take about 70 responses [3](https://generative-ai-newsroom.com/the-often-overlooked-water-footprint-of-ai-models-46991e3094b6).

The media ran with this estimate pretty hard. You started seeing headlines about ChatGPT “drinking” water with every interaction. Some publications even claimed that generating a 100-word email with GPT-4 could consume more than a full bottle of water (519 milliliters) [4](https://www.businessenergyuk.com/knowledge-hub/chatgpt-energy-consumption-visualized/).

Updated estimates for ChatGPT water usage

More recent data suggests those initial numbers were probably too high. Sam Altman from OpenAI recently said each ChatGPT query uses about 0.00032 liters of water – roughly one-fifteenth of a teaspoon [5](https://www.hindustantimes.com/technology/every-chatgpt-query-you-make-uses-water-and-sam-altman-has-revealed-the-exact-figure-101749632092992.html). So you’d need about 1,000 queries to use just 0.32 liters [5](https://www.hindustantimes.com/technology/every-chatgpt-query-you-make-uses-water-and-sam-altman-has-revealed-the-exact-figure-101749632092992.html).

Several researchers have pushed back on the original estimates too. One analysis suggests that when you factor in current model efficiency and typical conversation length, the actual water consumption is closer to 5ml per average conversation, not 500ml [1](https://www.seangoedecke.com/water-impact-of-ai/).

Inference vs training: which uses more water?

Here’s something most people don’t realize – training these models uses way more water than actually running them. Training GPT-3 alone probably consumed around 700,000 liters of water at Microsoft’s data centers [3](https://generative-ai-newsroom.com/the-often-overlooked-water-footprint-of-ai-models-46991e3094b6). For something like GPT-4, the water footprint could be about 10 times higher [1](https://www.seangoedecke.com/water-impact-of-ai/).

The water usage comes from three main sources: direct server cooling (which can evaporate 1-9 liters per kWh), electricity generation (averaging 7.6 liters per kWh), and manufacturing components like microchips (8-10 liters per chip) [6](https://cee.illinois.edu/news/AIs-Challenging-Waters).

Data centers globally already consume about 560 billion liters of water annually. Projections suggest this could more than double to 1,200 billion liters by 2030 as AI development ramps up[7](https://www.bloomberg.com/graphics/2025-ai-impacts-data-centers-water-data/).

How AI water use compares to your daily habits

Alright, so now let's see how these AI numbers actually stack up against stuff you do every day. Trust me, some of these comparisons are going to surprise you.

Showering and bathing

That 8-minute shower you take every morning? It uses about 60,000 milliliters of water[8](https://keryc.com/en/fact-check/ai-consumes-water-daily-habits-myth-reality-23r0PK). That's like asking ChatGPT 2,000 questions.

And if you're a bath person, filling up that tub takes around 136 liters [3](https://generative-ai-newsroom.com/the-often-overlooked-water-footprint-of-ai-models-46991e3094b6) – equivalent to over 4,500 AI queries. Even though about 90% of your shower water just goes straight down the drain[9](https://www.kjzz.org/the-show/2025-02-28/how-artificial-intelligence-uses-a-lot-of-water-and-why-thats-a-concern-for-states-like-arizona), you're still using way more water than any reasonable amount of ChatGPT usage.

Streaming video and music

Your digital entertainment habits add up too. Stream music for an hour and you're looking at about 250ml of water [10](https://www.warpnews.org/artificial-intelligence/ai-usage-has-less-environmental-impact-than-claimed-2/) – roughly 8 times what a single ChatGPT query uses.

That evening Netflix binge? One hour of video streaming or videoconferencing needs between 2-12 liters of water [2](https://nationalcentreforai.jiscinvolve.org/wp/2025/05/02/artificial-intelligence-and-the-environment-putting-the-numbers-into-perspective/). So your typical movie night probably uses more water than dozens of AI conversations combined.

Using social media and Zoom

Here's one that caught me off guard. Scrolling through social media for an hour uses approximately 430ml of water [10](https://www.warpnews.org/artificial-intelligence/ai-usage-has-less-environmental-impact-than-claimed-2/) – about 14 times more than asking ChatGPT a single question.

But the real kicker? A one-hour Zoom call consumes around 1,720ml of water [10](https://www.warpnews.org/artificial-intelligence/ai-usage-has-less-environmental-impact-than-claimed-2/). That's equivalent to about 57 ChatGPT queries. Remember, if you're asking ChatGPT five questions daily, that's only 150ml of water [2](https://nationalcentreforai.jiscinvolve.org/wp/2025/05/02/artificial-intelligence-and-the-environment-putting-the-numbers-into-perspective/) – way less than a single video call.

Washing clothes and dishes

Now we're getting to the heavy hitters. A single load of laundry uses approximately 117,000ml of water [8](https://keryc.com/en/fact-check/ai-consumes-water-daily-habits-myth-reality-23r0PK). To put that in perspective, that's equivalent to roughly 3,900 AI queries.

Washing dishes consumes around 23,000ml [8](https://keryc.com/en/fact-check/ai-consumes-water-daily-habits-myth-reality-23r0PK) – same as about 766 ChatGPT interactions. Even a single toilet flush uses 6,000ml of water [8](https://keryc.com/en/fact-check/ai-consumes-water-daily-habits-myth-reality-23r0PK), which equals 200 AI queries.

When you look at the big picture, the average person uses anywhere from 175-384 liters of water daily[8](https://keryc.com/en/fact-check/ai-consumes-water-daily-habits-myth-reality-23r0PK). Your ChatGPT usage barely shows up on that scale.

Why the numbers can be misleading

You've probably noticed that different articles give wildly different numbers for AI water usage. There's a good reason for that confusion – measuring this stuff accurately is actually pretty tricky.

Different models, different footprints

The huge differences in AI water usage estimates come down to researchers using completely different approaches [11](https://fas.org/publication/measuring-and-standardizing-ais-energy-footprint/). Some focus on counting AI queries, others look at hardware supply data [11](https://fas.org/publication/measuring-and-standardizing-ais-energy-footprint/). The models themselves vary massively too – a 1 MW data center might consume up to 25.5 million liters of water annually just for cooling [12](https://www.weforum.org/stories/2024/11/circular-water-solutions-sustainable-data-centers/), but smaller, more efficient models can use way less. The current metrics we use to measure data center efficiency often completely miss water consumption [11](https://fas.org/publication/measuring-and-standardizing-ais-energy-footprint/).

Location and time of day matter

Here's something that surprised me – where and when you run AI makes a huge difference. Training the same model uses less water in Iowa than Arizona because of climate differences [13](https://www.globalwaterforum.org/2023/11/02/a-double-edged-sword-ais-energy-water-footprint-and-its-role-in-resource-conservation/). It gets even more specific than that. Running AI operations at midday in California might be great for carbon efficiency due to solar power, but terrible for water efficiency because of the heat [14](https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology). So the "when" and "where" of AI usage can really change its water footprint [14](https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology).

Cooling systems and water recycling

The cooling tech makes a massive difference in water usage. Traditional cooling towers use drinking water and lose about 80% of it to evaporation [15](https://www.foodandwaterwatch.org/2025/04/09/artificial-intelligence-water-climate/). But newer tech like air-side economization can cut water usage by 80-90% compared to old methods [16](https://www.gigenet.com/blog/ai-water-consumption-the-hidden-environmental-cost-of-artificial-intelligence/). You've got closed-loop systems that recirculate water and immersion cooling that doesn't need water at all [16](https://www.gigenet.com/blog/ai-water-consumption-the-hidden-environmental-cost-of-artificial-intelligence/). Problem is, less than a third of data center operators actually track their water usage [12](https://www.weforum.org/stories/2024/11/circular-water-solutions-sustainable-data-centers/), so getting clear info about what companies are really doing is pretty challenging.

What this means for your AI usage

Now that you know where AI actually sits in your water usage, let's talk about what you can actually do about it. Your individual impact might seem tiny, but when millions of people are using these tools, it does add up.

Being mindful of unnecessary prompts

Each time you chat with ChatGPT, you're using water – roughly 500ml for a conversation of 20-50 questions and answers [14](https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology). So yeah, cutting down on the random "tell me a joke" prompts actually makes a difference. Before you hit send, just ask yourself if this query is actually useful to you.

Plus, being more specific with your questions not only saves water but gets you better answers anyway. Win-win.

Balancing AI use with other water-saving habits

Look, AI is just one piece of your environmental footprint. You don’t need to stop using ChatGPT entirely – that’s not realistic. What researchers are saying is that each ChatGPT request is like pouring out a bottle of water [17](https://medium.com/cictwvsu-online/the-environmental-cost-of-ai-why-efficient-prompts-matter-e038186133c3). But remember, your shower this morning used way more.

Instead of avoiding AI tools, I’d focus on supporting companies that are actually investing in better cooling technologies [18](https://www.linkedin.com/pulse/ai-use-water-consumption-conservation-strategies-chester-beard-cungc). And there are AI applications out there helping people save water elsewhere – smart irrigation systems, leak detection, stuff like that.

The role of transparency from tech companies

Here’s what’s frustrating – less than one-third of data center operators even track their water usage [19](https://www.whitecase.com/insight-our-thinking/ai-water-management-balancing-innovation-and-consumption). How are we supposed to make good choices when we don’t have the data? The US Government Accountability Office called this out too, saying AI systems are basically “black boxes” even to the people building them [20](https://news.slashdot.org/story/25/04/24/1556239/even-the-us-government-says-ai-requires-massive-amounts-of-water).

Things might change with regulations like the AI Environmental Impacts Act pushing for better reporting [6](https://cee.illinois.edu/news/AIs-Challenging-Waters). Until then, we’re kind of flying blind on the actual numbers.

Comparison Chart

||
||
Activity	**Water Consumption**	**Equivalent AI Queries (Based on 5-30ml/query)**
1 ChatGPT query	5-30 ml	1
Streaming Music (1 hour)	250 ml	8 – 50
Social Media (1 hour)	430 ml	14 – 86
Zoom Call (1 hour)	1,720 ml	57 – 344

Toilet Flush	6,000 ml	200 – 1200
Streaming Video/Video Conferencing (1 hour)	2,000 -12,000 ml	67 – 2400
Washing Dishes	23,000 ml	766 – 4600
8-minute Shower	60,000 ml	2,000 – 12,000
Load of Laundry	117,000 ml	3,900 – 23,400
Bath	136,000 ml	4,533 – 27,200

Conclusion

Look, most of the conversation around AI water usage misses the bigger picture. Yeah, AI systems use water – that part’s true. But when you actually compare it to your daily routine, it’s pretty small. Those five ChatGPT queries you might do each day? 150ml of water. Your 8-minute shower? 60,000ml. Even a single Zoom call uses 1,720ml.

Don’t get me wrong – I’m not saying we should ignore AI’s environmental impact completely. Training these big models like GPT-4 takes serious resources, and data centers keep using more water every year. But getting worked up about your ChatGPT usage while not thinking about your shower habits doesn’t make much sense.

The real issue here is that tech companies aren’t being transparent about this stuff. You can’t make good choices about your AI use when you don’t have the actual numbers. Less than a third of data center operators even track their water usage. That needs to change.

For most of us, the math is pretty straightforward. Your morning shower or washing machine uses way more water than asking AI questions. Still, being mindful about all your habits – digital and otherwise – is part of being responsible.

I think the key is keeping perspective. AI can be incredibly useful when you use it thoughtfully. Understanding what it actually costs helps you make better decisions instead of just reacting to scary headlines. Next time you use ChatGPT, you’ll know that your coffee probably has a bigger water footprint.

The tech companies need to do better with transparency and efficiency. But in the meantime, you don’t need to feel guilty about using these tools when they genuinely help you get things done.

References

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Comments

u/Rainjewelitt4211 • 0 points • 2026-04-04 03:52 UTC

yes, training GPT-3 used a ton of water but your daily usage with five queries is honestly roughly 150ml while your morning shower is 60,000ml so the math isn't close. the actual problem is that less than a third of data center operators even track their water usage, so we're debating estimates while the companies responsible aren't being held to any reporting standard

u/Exact_Wolverine_2341 • 1 points • 2026-04-01 12:19 UTC

I don't support AI art unless you tell everyone that it's AI art and dont try to claim it as "your own", but I'm seriously tired of people blaming AI for the bad environment...

> **u/Mikhael_Love** • 1 points • 2026-04-01 13:32 UTC
>
> >unless you tell everyone that it's AI
>
> Really, the 'tell everyone' approach is intended to cater to people's icky feelings. There is no need for a blanket 'everything AI must be labeled' mandate.
>
> I asuume by "your own" you mean fraudulently claiming something that is not true.
>
> I wrote a comment about my thoughts on this some time ago, so if you are interested in reading the whole thing let me know and I try to dig it up for you.

u/Lopsided_Biscotti_31 • 0 points • 2026-01-27 22:00 UTC

Because there's no reason for ai and the extra water it uses… ai adds nothing new to what we can already do, just makes it faster

u/Goober1082 • 1 points • 2026-01-21 04:08 UTC

Rats

u/[deleted] • 1 points • 2026-01-19 15:55 UTC

[removed]

u/[deleted] • 1 points • 2025-08-25 20:00 UTC

[removed]

u/NarukamiOgoshoX • 1 points • 2025-08-11 12:01 UTC

And whenever they say pick up a pencil, they never think about how pencils are made of wood meaning created from trees and that basically means

Pencils are destroying the environment while AI isn't doing anything to harm environment.

Then again... The Antis may be the same people from the "Stop Oil" protest, including the two girls that threw campbell soup at a painting.
It was just stupid, even more so when the thing has protective glass so it was easily wiped off lol

> **u/onlyaskingquestionss** • 1 points • 2026-04-01 06:24 UTC
>
> That's not entirely true ai centers consume massive amounts of electricity which is often gained via the use of carbon emissions, although to solely point the blame on AI is sensationalist. The larger issue is that it uses municipal water, which drives up water costs as a whole

u/GlitteringTravel6112 • 3 points • 2025-08-11 02:51 UTC

LMAO i saw someone yesterday try to claim that EVERY SINGLE GENERATION USED 700,000 gallons of water!

> **u/[deleted]** • 1 points • 2025-08-11 07:16 UTC
>
> [removed]

u/Early-Dentist3782 • 0 points • 2025-08-10 23:18 UTC

We know

u/adamkad1 • 3 points • 2025-08-10 18:27 UTC

And they also come up with that argument that the water could be given to those who need it... Yeah, im sure they'd thank them while dying from all sorts of things that would probably happen if they drank the non drinkable water most ai stuff uses

> **u/Seeker_Of_Hearts** • 1 points • 2025-08-11 13:25 UTC
>
> Moreover, it wouldn't have been given to them even if AI wasn't a thing, it'd just be used for something else.. Places that lack water existed before AI, it's not like they've been given 'extra water' by anyone.
>
> It's the same logic a nephew of mine used on me when I ended up not buying something I wanted to, he said "you were gonna use the money anyway so just give it to me" (He is saving up for a console, for his age, he really really needs the money)

> **u/Mikhael_Love** • 3 points • 2025-08-10 18:31 UTC
>
> This is a good point. Many use treated effluent, reclaimed water and greywater.

u/Sad_Pineapple5909 • 1 points • 2025-08-10 12:55 UTC

Mom hurry up a new PhD thesis on defending AI has dropped.

u/Person012345 • 11 points • 2025-08-10 12:44 UTC

Post it in AI wars too I'd say, more likely people will see it there.

u/king-amgh • 3 points • 2025-08-10 05:25 UTC

You're saying that the AI water usage includes chip manufacturing, so it would be fair to also include the water usage of making the 2-hour movie you're watching from Netflix.